Sahil Gulania

Curriculum Vitae

Department of Chemistry
University of Southern California
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Personal Information

Date of Birth **1 March 1994**.

Education

2016–present **Doctor of Philosphy, Chemistry**, *University of Southern California (USC)*, Los Angeles, USA.

2011–2016 **Integrated Master of Science (Chemistry)**, *National Institute of Science Education and Research (NISER)*, Bhubaneswar, India.

Ph.D Research

Electronic structure theory development and application: Equation of Motion Coupled Cluster method

Duration : 2016-present
Adviser : Dr. Anna Krylov, Professor, USC, Los Angeles, USA

Publications (published)

2020

- **Limitations of Hartree-Fock with quantum resources**
  Sahil Gulania, and James Daniel Whitfield
  arXiv:2007.09806 (2020)

- **Two cycling centers in one molecule: Communication by through-bond interactions and entanglement of the unpaired electrons**
  Maxim Ivanov, Sahil Gulania, and Anna I. Krylov
  J. Phys. Chem. Lett. 11, 1297 – 1304 (2020)

- **The quest to uncover the nature of benzonitrile anion**
  Sahil Gulania, Thomas-C. Jagau, Andrei Sanov, and Anna I. Krylov
  Phys. Chem. Chem. Phys. 22, 5002 – 5010 (2020)

- **Domain-Specific Compilers for Dynamic Simulations of Quantum Materials on Quantum Computers**
  Lindsay Bassman, Sahil Gulania, Connor Powers, Rongpeng Li, Thomas Linker, Kuang Liu, T. K. Satish Kumar, Rajiv K. Kalia, Aiichiro Nakano and Priya Vashishta
  arXiv:2004.07418 (2020)

2019

- **Young frames for quantum chemistry**
  Sahil Gulania and James Daniel Whitfield
  arXiv:1904.10469 (2019)

- **EOM-CC guide to Fock-space travel: The C_2 edition**
  Sahil Gulania, Thomas-C. Jagau and Anna I. Krylov
  Faraday Disc. 217, 514–532 (2019), DOI:10.1039/c8fd00185e

2018

- **Bound and continuum-embedded states of cyanopolyyne anions**
  Wojciech Skomorowski, Sahil Gulania, and Anna I. Krylov
  Phys. Chem. Chem. Phys., 20, 4805-4817 (2018) , DOI:10.1039/C7CP08227D

2017

- **Channel branching ratios in CH_2CN^- photo detachment: Rotational structure and vibrational energy redistribution in auto-detachment**
  Justin Lyle, Olivia Wedig, Sahil Gulania, Anna I. Krylov, and Richard Mabbs
  J. Chem. Phys. 147, 234309 (2017), DOI:10.1063/1.5001475

Publications (in-preparation)

- **Development and benchmark of EOM-DEA(Equation of motion - double electron attachment) theory for diradicals and two-electron excited states.**

Scholarships and Academic Awards

- Dornisfe Graduate School Fellowship for 5-years of study at USC.
- Innovation in Science Pursuit for Inspired Research (INSPIRE) granted by Ministry of Human Resource Development (MHRD) for all 5-years of study, INDIA
- Scholarship for Higher Education from CBSE (ISPIRE) for being in top 1 % in Central Board of Secondary Education (Class XII), INDIA

Master of Science Thesis

Implementation of Stationary Point Search and Intrinsic Reaction Coordinate following algorithms for Force Modified Potential Energy Surface

Duration : 2014-2016

Project Guide : Dr. Upakarasamy Lourderaj, Reader-F, NISER, Bhubaneswar